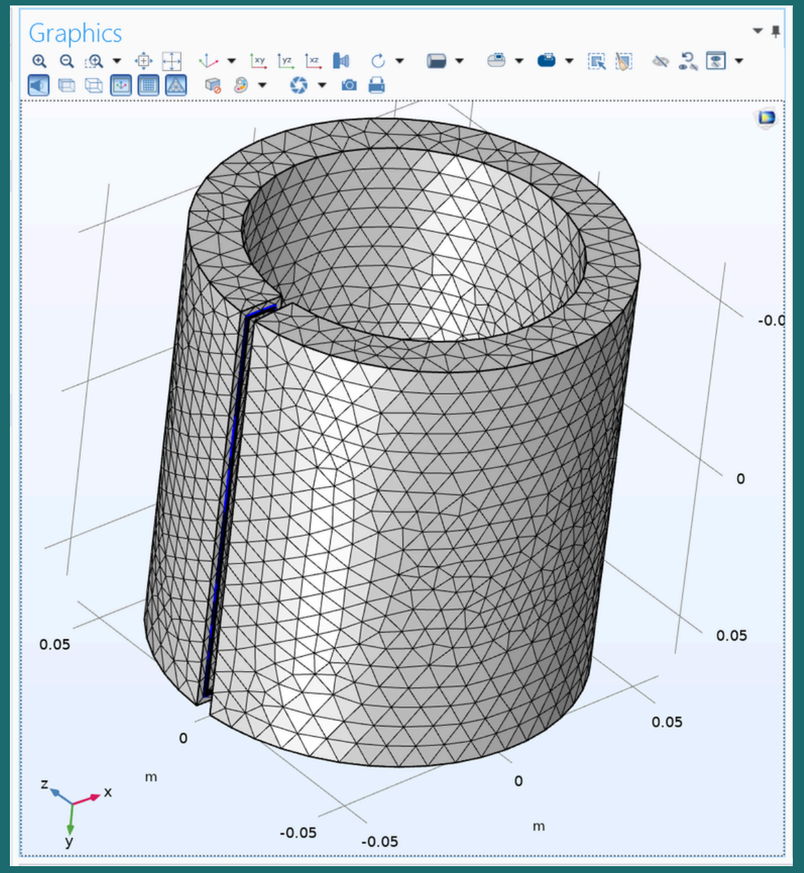
Cooling Jacket



Goals/ Constraints

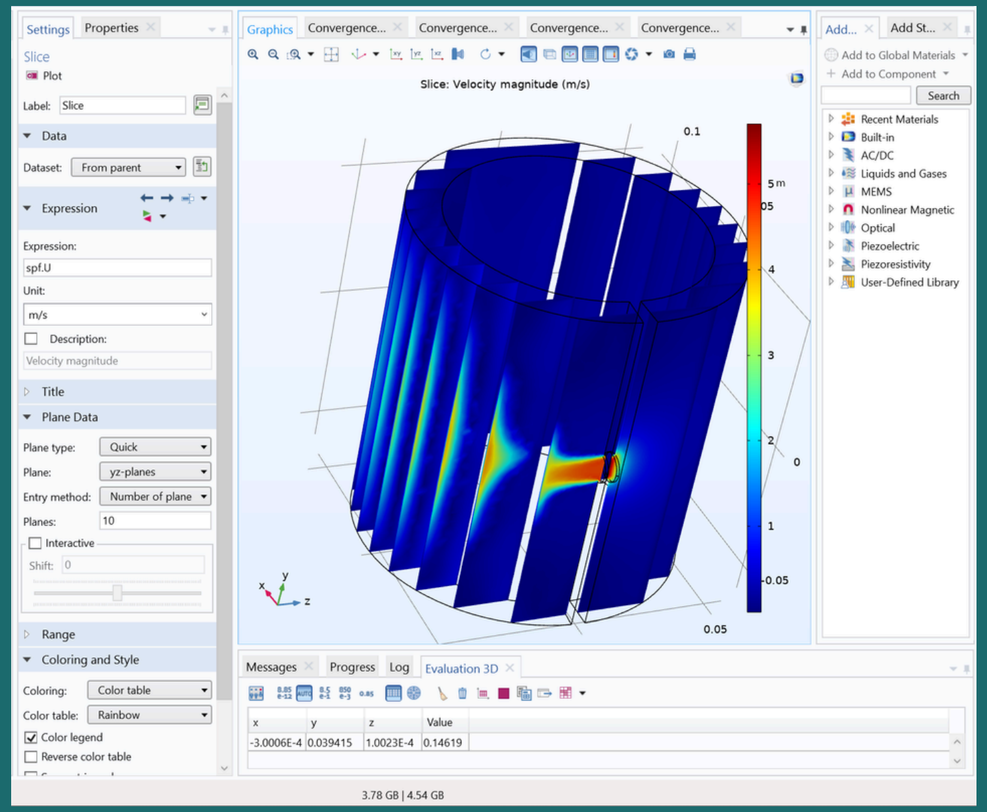
1. Dissipate 2000W of heat while maintaining motor temps well below 120 °C
2. Keeps total cooling system pressure drop < 1.5 bar at 10lpm (ideal flow rate for MC cooling according to datasheet)
3. Limit water temp rise across jacket to < 5 ° C
4. Maximise ease of assembly, routing, etc.
5. Design for CNC-based manufacturing.

Possible Architectures



1. Single Pass

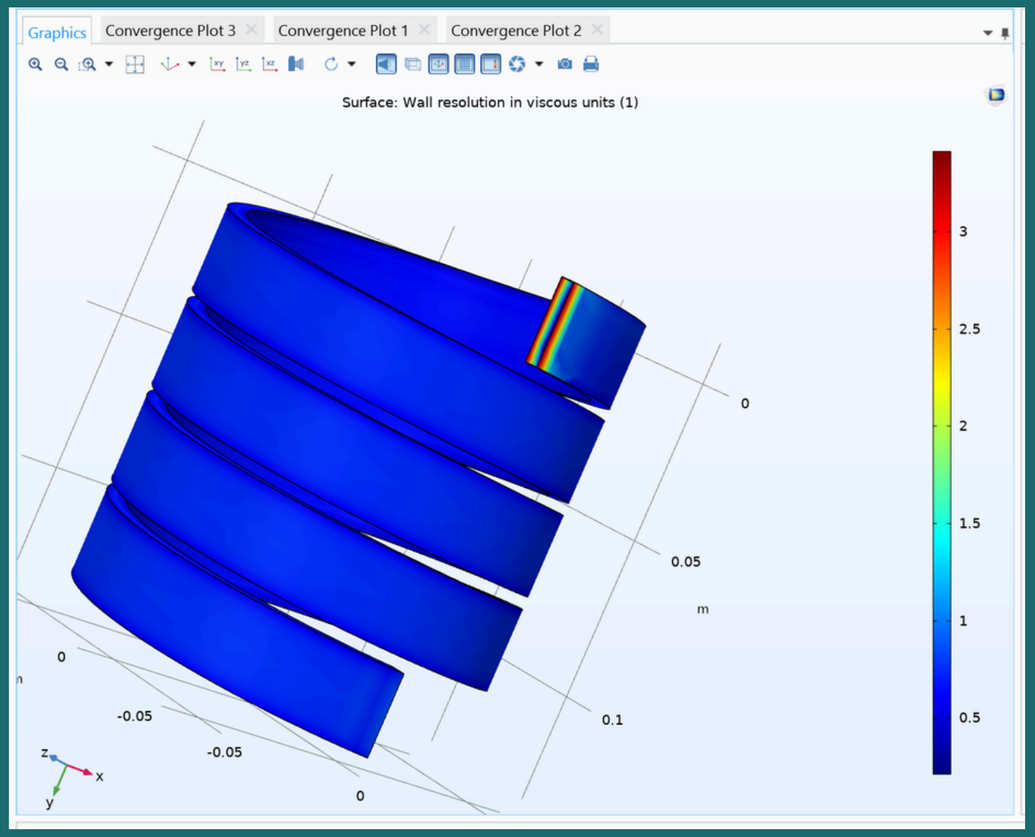
- Easiest to manufacture
- Very low pressure drop
- Not very good HTC unless channel height is approx 1mm



- The water velocity profile is highly non-uniform, with values close to 0 over a large area.
- The issue could be solved by adding multiple inlets/outlets along the length, but the design becomes complex and essentially resembles the helical design.

2. Helical Channels

- Good balance of HTC and pressure drop
- Can be CNC’ed
- Channel dimensions can be adjusted and optimised for ideal flow conditions within our constraints



3. ZigZag Channel

- Similar HTC to helical
- Significantly higher pressure drop due to repeated change of flow direction
- Channel dimensions can be adjusted and optimised for ideal flow conditions within our constraints

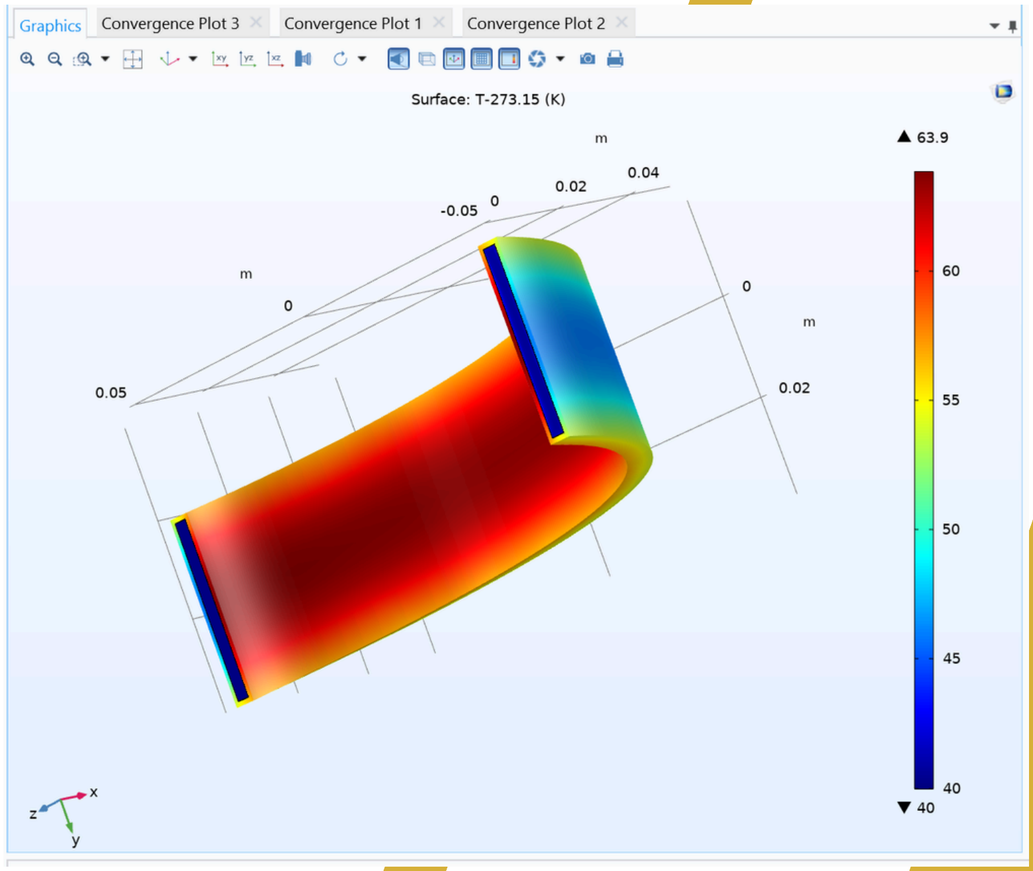
2		Helical flat	Single Turn	Zig Zag
3	approx peak surface temp (K)	324.00	395.00	328.00
4	approx HTC (W/m^2K)	4400.60	583.34	3215.25
5	Pressure drop (bar)	0.15	0.01	0.20
6	Manufacturability	medium	Easy	Medium

Optimising Helical Geometry

- Took initial guess for channel dimensions based on inspecting other teams' designs that were running the same/similar motors.

- Ran simulations for channel heights of 3mm, 4.5mm, and 6mm for jackets with 3, 4, and 5 turns (corresponding to channel widths of 36, 26 and 20mm, respectively)

- Sim was run on a half-turn section to reduce computation time and get quicker results for qualitative comparisons between geometries.



Values of parameters

- Turbulence Model: SST

1. For accurate results, the wall resolution near walls should be close to 1.

2. After applying the mesh and running the simulation, the final value of y1 can be iterated on based on the wall resolution plot

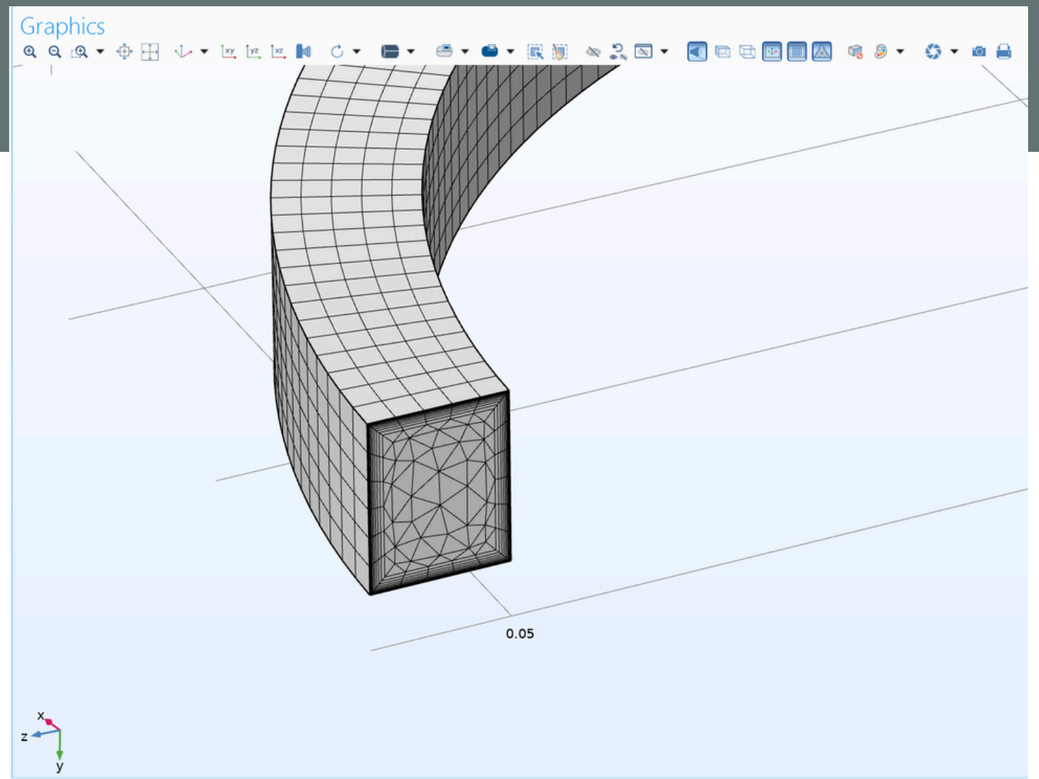
$$y^+ = \frac{u_\tau \, y_1}{\nu}$$

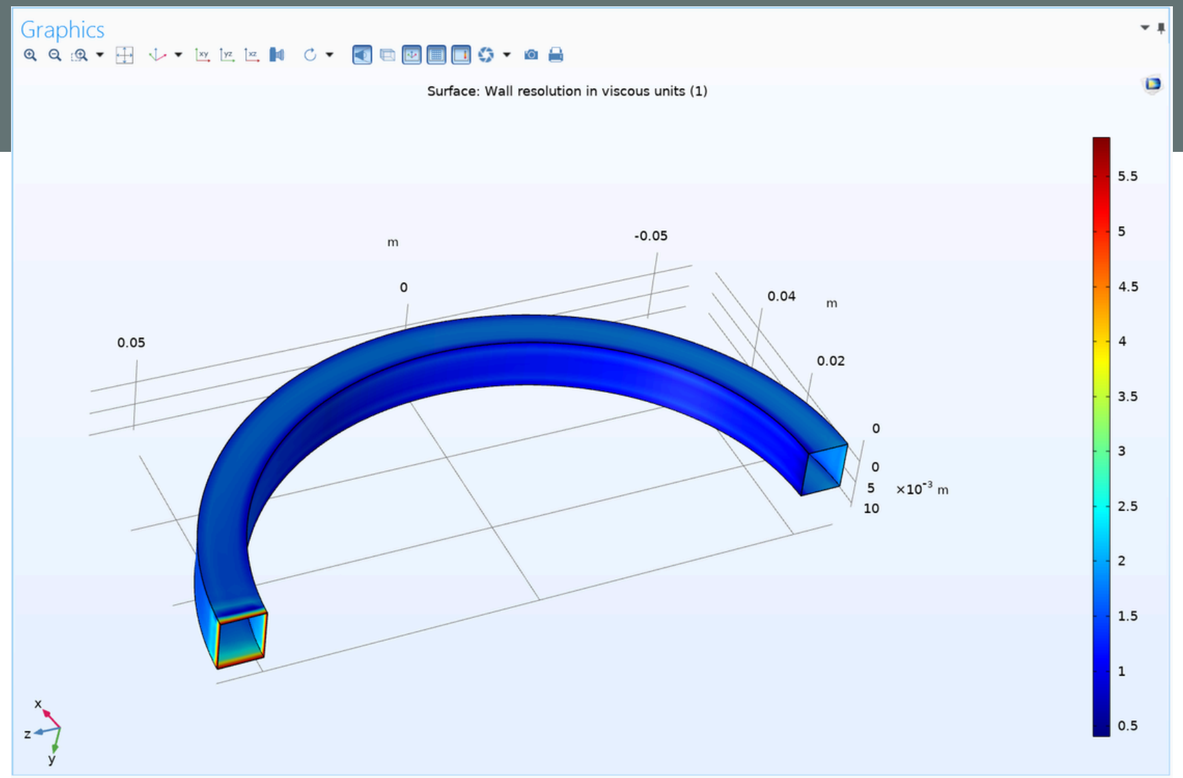
Where y1 is the first layer height in the boundary layer mesh and ut is the friction velocity, given by

$$u_\tau = U_\infty \sqrt{\frac{C_f}{2}}$$

Where Cf is the friction factor, which can be obtained from empirical relations and Uinf is the characteristic velocity for the case.

- After deciding the 1st-layer height, choose the expansion factor and the number of layers so that approx. 20% (increasing to 40% for more viscous fluids) of the channel area is covered by the boundary-layer mesh.





- Running simulations for the 9 geometries mentioned earlier yielded the following results.
- Note: for these initial sims, the boundary conditions I set added a lot of extra heat to the water and jacket, so the temperature values (temp rise) are higher than expected. But since the same conditions were applied to all sims, the data still gives a qualitative comparison of the geometries.
- Went on to choose new geometries

HTC	3 turns	4 turns	5 turns
3mm	2735.86	6356.86	8105
4.5mm	2315.71	2759.15	3242
6mm	2026.25	2446.79	2819.13

pressure	3 turns	4 turns	5 turns
3mm	0.09	0.15	0.3
4.5mm	0.03	0.06	0.12
6mm	0.018	0.035	0.07

surface temp	3 turns	4 turns	5 turns
3mm	63.7	50.2	48
4.5mm	68	63.5	60
6mm	72	66.5	63

temp rise	3 turns	4 turns	5 turns
3mm	3	5.5	11
4.5mm	2.7	2.5	2.6
6mm	2.1	2.4	2.3

Optimising Helical Geometry

- I moved forward with the 4-turn, 3mm geometry (26x3mm) until I ran into an issue.
- A low number of turns means a large helix pitch, putting any geometry with a large pitch on the motor would mean large zones of it (near the inlet and outlet) are not in contact with the flow.
- Started trying more turns (lower width) and larger channel heights. Moved to 7 -11 turns with heights of 5 - 10mm.

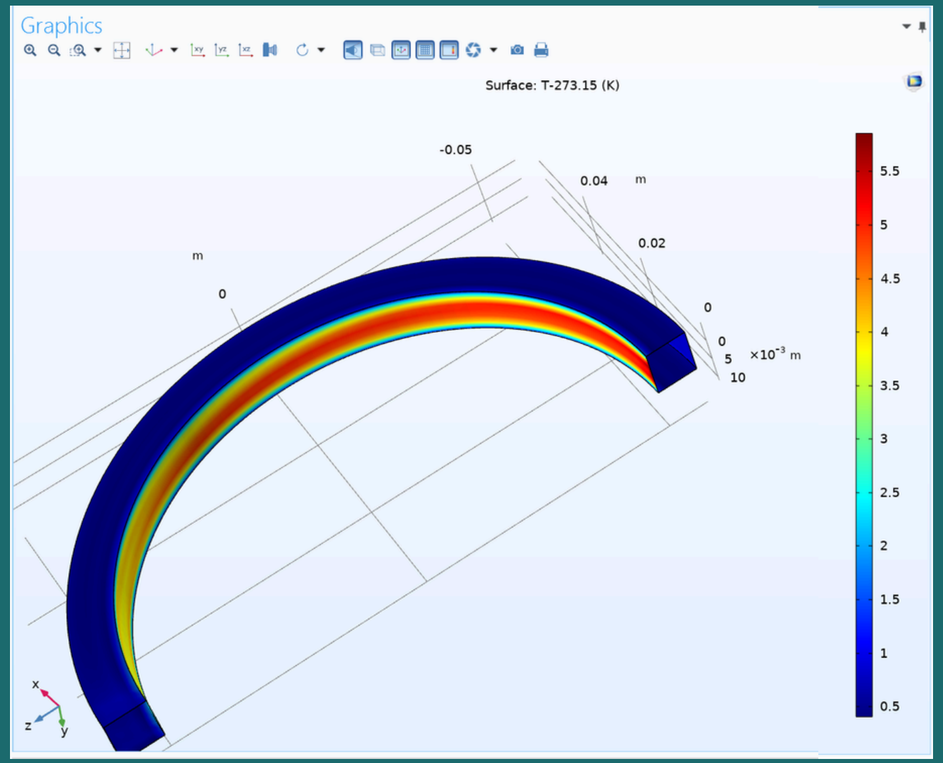
Selected optimal geometry of the channel based on the new data now.

Flowrate : 10 lpm Heat flux : 2167W									
HTC	9 turns	11 turns	7 turns		surface temp	9 turns	11 turns	7 turns	
6mm	13637.19	16033.67	8551.47		6mm	44.68	43.98	45.58	
8mm	11852.27	13621.77	9298.86	5mm	8mm	45.39	44.69	45.13	5mm
10mm	10462.06	12271.15			10mm	46.10	45.20		
pressure	9 turns	11 turns	7 mm		HTC/kPa	9 turns	11 turns		
6mm	0.61	1.13	0.3		6mm	223.560549	141.8908557	285.0489599	
8mm	0.25	0.56	0.49	5mm	8mm	474.0908855	243.2458793	189.7725553	5mm
10mm	0.16	0.3			10mm	653.8786048	409.0384696		

11 turns, 10mm was chosen

Corrected boundary conditions :

- Convection with air on jacket outer surface,
- Motor losses heat flux (48980 W/m2) on jacket inner surface
- Same heat flux on water inner surface
- Open boundary on all 4 walls of water, with upstream temperatures defined based on guesses of jacket surface temperature. (iterated on and improved after running the sim)
- Internal forced convection on channel walls, with diameter = 2LB/(L+B)
- Pressure at outlet set as 1 atm (arbitrary)



Validation

- Once the simulation is run, verify that the results are physical, i.e., check the energy balance
- Check that mcpdt equals the supplied heat.
- To find dt, use the velocity-weighted average temperature of the outlet.
- Basically, do a surface integration of u*T over the outlet surface, then an integration of u, and take the ratio.

$$T_{out} = \frac{\int \rho u_n T \, dA}{\int \rho u_n \, dA}$$

Velocity-weighted average temperature

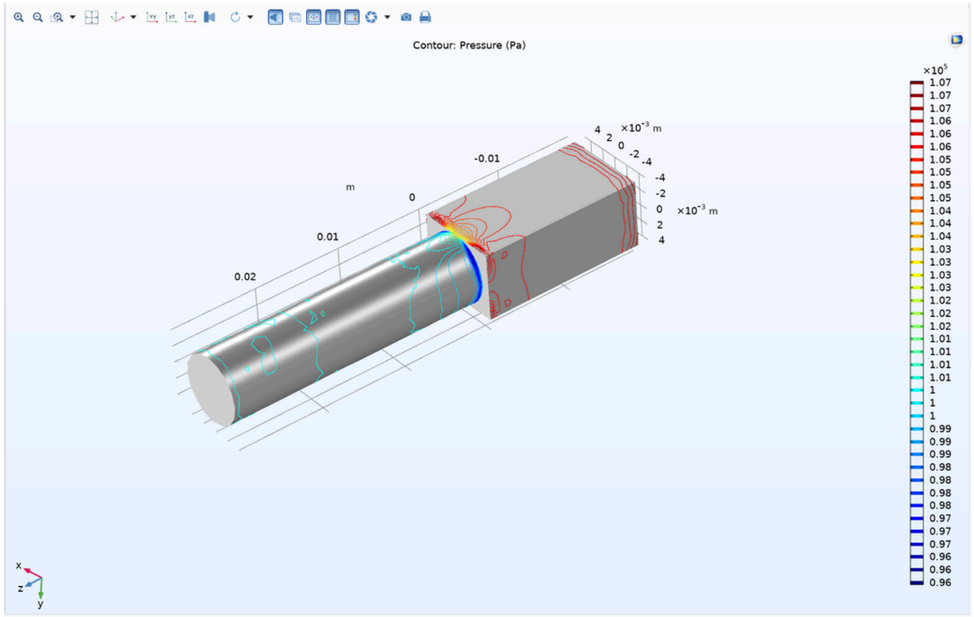
- Simulation results of m*cp*dt were within 2-5% of theoretical values.
- For a half-turn sim, the water temperature can be assumed constant at 40 °C (it actually varies by only about 0.2 °C).
- Thus, the HTC between the jacket's inner surface and the water is approximately

$$h = q''/(T_s - T_w)$$

Where Ts is the jacket's inner surface temperature.

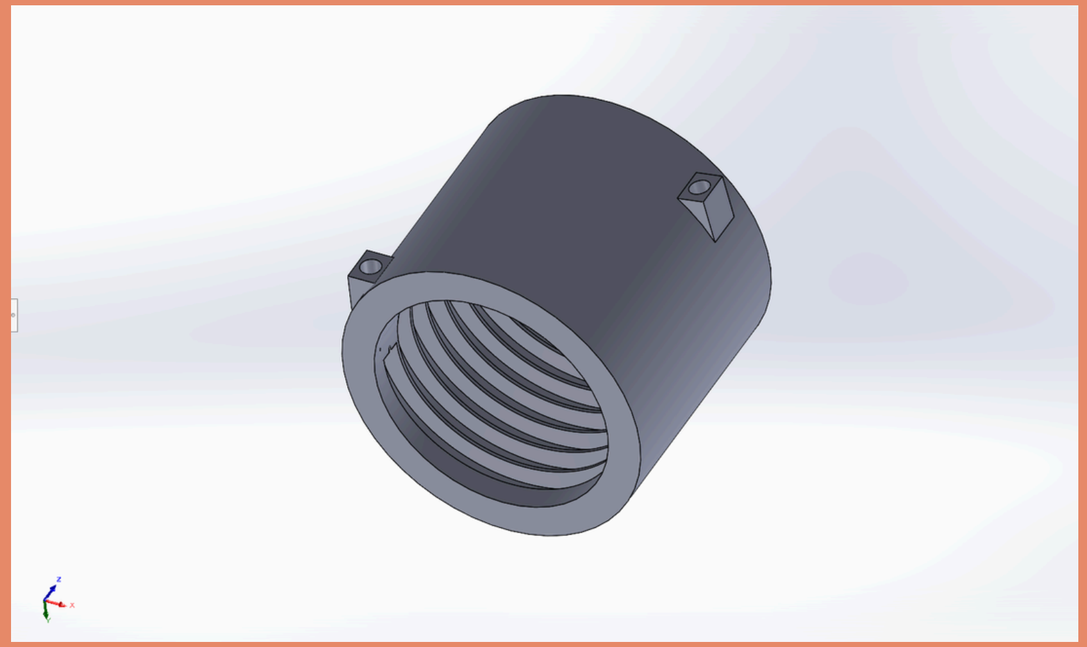
Inlet-Outlet design

- The inlet and outlet diameters were kept at 8mm (because the flow cross-section was 8 × 10), with an outer dimension of 11 × 15 mm.
- We checked the inlet/outlet pressure using a simple simulation of these cross sections.
- The pressure drop across the inlet/outlet was 0.05 bar.



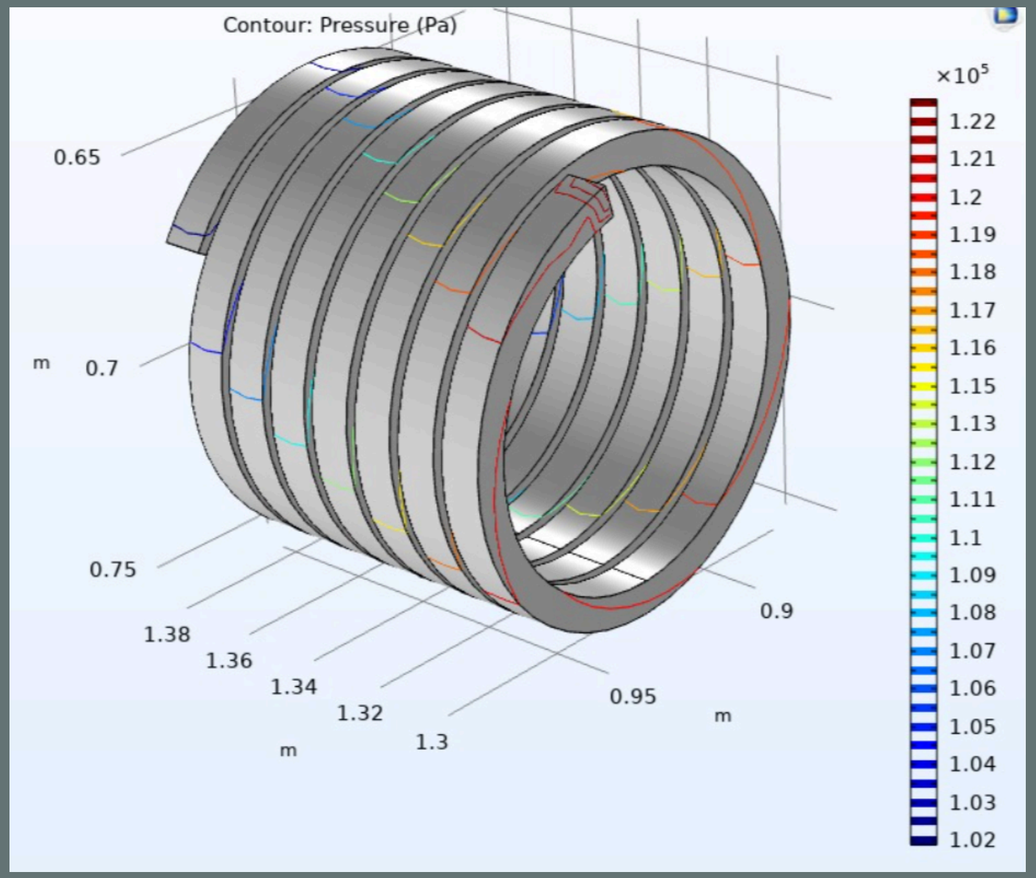
Finalising CAD design

- The inner surface of the cooling surface was cut-swept with dimensions of 8*10 and 8.4 revolutions at a pitch of 10.04.
- The distance from the end of the inlet outlet to the end faces is approximately the same, around 6-7 mm.
- 8.4 revolutions were chosen so that the inlet and outlet align in the same direction, which would give ease in the handling of the assembly.



- Comsol simulation results on this fluid section were as follows:

Params	**Values**
Pressure drop	0.227 bar
Surface temp	45.2
Flow rate	10 lpm
HTC	11.8k



Thank you